AstroCLIMB system study

Auditable Multi-View Hosted Multimodal Judging for AstroCLIMB

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Hosted judging improves local validation.
Fixed fusion does not beat Luna-only.

1 Task and leakage audit

Classify relationships between astronomy figure/caption endpoints across image-image, image-caption and caption-caption pairs.

SF Same figure
SP Same paper
RP Related papers
UP Unrelated papers

8,000 training pairs and 2,000 validation pairs
Zero endpoint or pair overlap in the frozen split

> **Observable-content components only.**
> **The split is not paper-disjoint.**

2 Fixed six-call protocol

Content-only inputs: raw captions and preprocessed images. No separate DOI, title, author or citation-metadata fields reach the judge.

Direct	Counterfactual	Boundary
A/B + B/A	A/B + B/A	A/B + B/A

GPT-5.6 Luna (medium reasoning effort)

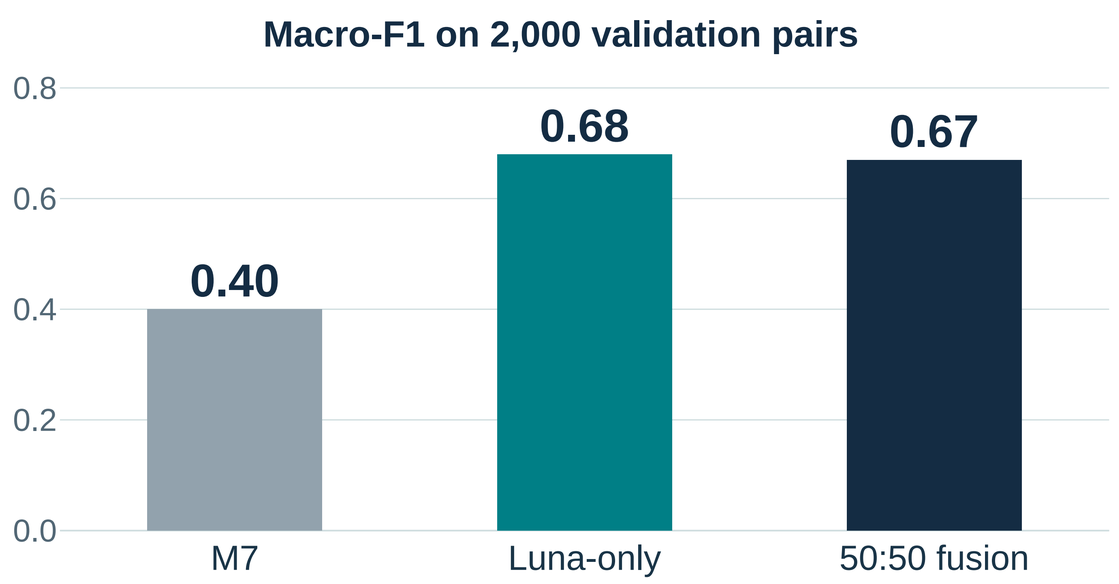
Six valid probability vectors → **Geometric mean**

Normalize probabilities and mask impossible SF labels. Fuse Luna and the masked M7 baseline with fixed 50:50 weights.

All six calls succeed, or fall back to M7

Reported run: 12,000 / 12,000 successful calls.
Append-only cache, anonymous keys and digest-bearing receipts support inspection.

3 Frozen validation results



Fusion vs M7: +0.27 macro-F1

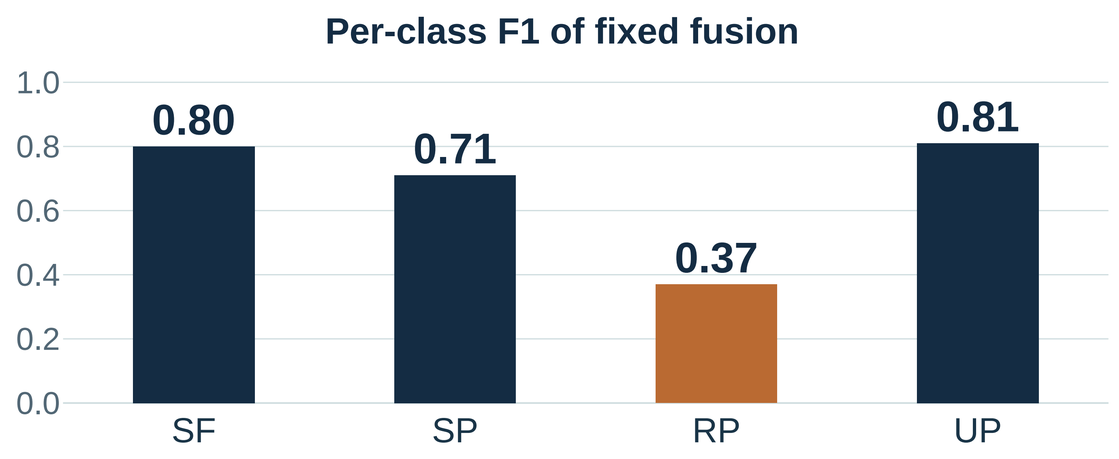
95% CI [+0.24, +0.29]

Paired component bootstrap, 2,000 replicates, conditional on the fixed cached run.

Fusion is a negative result

The prespecified fusion scores 0.011 below Luna-only. No fusion weight was retuned, and no significance claim is made for this difference.

4 Related papers remain hardest



263 of 600 related-paper pairs are classified as same paper

The prompt uses substantive relatedness as a proxy for the benchmark's citation link. This mismatch is a plausible contributor, not an established cause.

Scope of the evidence

One local split and cached service run. No official Luna score or paper-level generalization.
Conditional intervals exclude API randomness and serving drift. Pretraining contamination remains unknown.

Paper: openreview.net/forum?id=kt7YLPEj6p